Specification Grid – 2081

First Terminal Examination

Class : 10 Subject : Science and Technology F.M. : 50Time : 2 Hours

<table>
<tr><th rowspan="2">S.N.</th><th rowspan="2">Unit</th><th rowspan="2">Working Hour</th><th rowspan="2">Group wise Marks</th><th rowspan="2">Unit wise Marks</th><th colspan="4">Cognitive Level</th></tr>
<tr><th>K
15 %</th><th>U
30 %</th><th>A
30 %</th><th>HA
25 %</th></tr>
<tr><td>1</td><td>Scientific Learning</td><td>5</td><td>7</td><td>7</td><td rowspan="7">MCQ
1×1

V Short
1×1

Short
1×2

Long
1×4</td><td rowspan="7">MCQ
2×1

V Short
1×1

Short
2×2

Long
2×4</td><td rowspan="7">MCQ
1×1

V Short
2×1

Short
4×2

Long
1×4</td><td rowspan="7">MCQ
3×1

V Short
1×1

Short
2×2

Long
1×4</td></tr>
<tr><td>2</td><td>Classification of Organisms</td><td>9</td><td rowspan="2">14</td><td>9 ± 2</td></tr>
<tr><td>3</td><td>Lifecycle of Honeybee</td><td>4</td><td>5 ± 2</td></tr>
<tr><td>7</td><td>Force and Motion</td><td>10</td><td rowspan="2">16</td><td>9 ± 2</td></tr>
<tr><td>8</td><td>Pressure</td><td>5</td><td>7 ± 2</td></tr>
<tr><td>14</td><td>Classification of Elements</td><td>9</td><td rowspan="2">13</td><td>8 ± 2</td></tr>
<tr><td>15</td><td>Chemical Reaction</td><td>6</td><td>5 ± 2</td></tr>
<tr><td colspan="2">Total</td><td>48</td><td>50</td><td>50</td><td>8</td><td>15</td><td>15</td><td>12</td></tr>
</table>

S.N.	Questions Type	Marking Schedule	Total
1	Multiple Choice Question (MCQ)	7 ×1=7	7
2	Very Short Questions (VS)	5 ×1=5	5
3	Short Questions (S)	9×2=18	18
4	Long Questions (L)	5×4=20	20
	Total	50	50

Specification Grid – 2081

Second Terminal Examination

Class : 10 Subject : Science and Technology F.M. : 50Time : 2 Hours

<table>
<tr><th rowspan="2">S.N.</th><th rowspan="2">Unit</th><th rowspan="2">Working Hour</th><th rowspan="2">Group wise Marks</th><th rowspan="2">Unit wise Marks</th><th colspan="4">Cognitive Level</th></tr>
<tr><th>K
15 %</th><th>U
30 %</th><th>A
30 %</th><th>HA
25 %</th></tr>
<tr><td>1</td><td>Scientific Learning</td><td>5</td><td rowspan="2">7</td><td>3 ± 2</td><td rowspan="13">MCQ
1×1

V Short
1×1

Short
1×2

Long
1×4</td><td rowspan="13">MCQ
2×1

V Short
1×1

Short
2×2

Long
2×4</td><td rowspan="13">MCQ
1×1

V Short
2×1

Short
4×2

Long
1×4</td><td rowspan="13">MCQ
3×1

V Short
1×1

Short
2×2

Long
1×4</td></tr>
<tr><td>13</td><td>Information and Communication Technology</td><td>10</td><td>4 ± 2</td></tr>
<tr><td>2</td><td>Classification of Organisms</td><td>9</td><td rowspan="4">14</td><td>3 ± 2</td></tr>
<tr><td>3</td><td>Lifecycle of Honeybee</td><td>4</td><td>2 ± 2</td></tr>
<tr><td>4</td><td>Heredity</td><td>16</td><td>5 ± 2</td></tr>
<tr><td>5</td><td>Physiological Structure and Life Process</td><td>12</td><td>4 ± 2</td></tr>
<tr><td>7</td><td>Force and Motion</td><td>10</td><td rowspan="4">16</td><td>6 ± 2</td></tr>
<tr><td>8</td><td>Pressure</td><td>5</td><td>5 ± 2</td></tr>
<tr><td>9</td><td>Heat Energy</td><td>10</td><td>5 ± 2</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>14</td><td>Classification of Elements</td><td>9</td><td rowspan="4">13</td><td>3 ± 2</td></tr>
<tr><td>15</td><td>Chemical Reactions</td><td>6</td><td>2 ± 2</td></tr>
<tr><td>16</td><td>Some Gases</td><td>8</td><td>5 ± 2</td></tr>
<tr><td>17</td><td>Metals</td><td>5</td><td>3 ± 2</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="2">Total</td><td>124</td><td>50</td><td>50</td><td>8</td><td>15</td><td>15</td><td>12</td></tr>
</table>

S.N.	Questions Type	Marking Schedule	Total
1	Multiple Choice Question (MCQ)	7 ×1=7	7
2	Very Short Questions (VS)	5 ×1=5	5
3	Short Questions (S)	9×2=18	18
4	Long Questions (L)	5×4=20	20
	Total	50	50

Specification Grid – 2081

Third Terminal Examination and Pree SEE Examination

Class : 10 **Subject : Science and Technology** **F.M. : 75Time : 3 Hours**

<table>
<tr><th rowspan="2">S.N.</th><th rowspan="2">Unit</th><th rowspan="2">Working Hour</th><th rowspan="2">Group wise Marks</th><th rowspan="2">Unit wise Marks</th><th colspan="4">Cognitive Level</th></tr>
<tr><th>K
15 %</th><th>U
30 %</th><th>A
30 %</th><th>HA
25 %</th></tr>
<tr><td>1</td><td>Scientific Learning</td><td>5</td><td rowspan="2">10</td><td>3 ± 2</td><td rowspan="19">MCQ
1×1

V
Short
1×1

Short
3×2

Long
1×4</td><td rowspan="19">MCQ
3×1

V Short
3×1

Short
4×2

Long
2×4</td><td rowspan="19">MCQ
3×1

V
Short
3×1

Short
4×2

Long
2×4</td><td rowspan="19">MCQ
3×1

V Short
2×1

Short
3×2

Long
2×4</td></tr>
<tr><td>13</td><td>Information and Communication Technology</td><td>10</td><td>7 ± 2</td></tr>
<tr><td>2</td><td>Classification of Organisms</td><td>9</td><td rowspan="5">19</td><td>3 ± 2</td></tr>
<tr><td>3</td><td>Lifecycle of Honeybee</td><td>4</td><td>2 ± 2</td></tr>
<tr><td>4</td><td>Heredity</td><td>16</td><td>6 ± 2</td></tr>
<tr><td>5</td><td>Physiological Structure and Life Process</td><td>12</td><td>5 ± 2</td></tr>
<tr><td>6</td><td>Nature and Environment</td><td>7</td><td>3 ± 2</td></tr>
<tr><td>7</td><td>Force and Motion</td><td>10</td><td rowspan="6">26</td><td>4 ± 2</td></tr>
<tr><td>8</td><td>Pressure</td><td>5</td><td>3 ± 2</td></tr>
<tr><td>9</td><td>Heat Energy</td><td>10</td><td>5 ± 2</td></tr>
<tr><td>10</td><td>Wave</td><td>15</td><td>6 ± 2</td></tr>
<tr><td>11</td><td>Electricity and Magnetism</td><td>12</td><td>5 ± 2</td></tr>
<tr><td>12</td><td>Universe</td><td>5</td><td>3 ± 2</td></tr>
<tr><td>14</td><td>Classification of Elements</td><td>9</td><td rowspan="6">20</td><td>4 ± 2</td></tr>
<tr><td>15</td><td>Chemical Reactions</td><td>6</td><td>2 ± 2</td></tr>
<tr><td>16</td><td>Some Gases</td><td>8</td><td>5 ± 2</td></tr>
<tr><td>17</td><td>Metals</td><td>5</td><td>3 ± 2</td></tr>
<tr><td>18</td><td>Hydrocarbon and Its Compound</td><td>6</td><td>3 ± 2</td></tr>
<tr><td>10</td><td>Chemicals Used in Daily Life</td><td>6</td><td>3 ± 2</td></tr>
<tr><td colspan="2">Total</td><td>160</td><td>75</td><td>75</td><td>12</td><td>22</td><td>22</td><td>19</td></tr>
</table>

S.N.	Questions Type	Marking Schedule	Total
1	Multiple Choice Question (MCQ)	10×1=10	10
2	Very Short Questions (VS)	9×1=9	9
3	Short Questions (S)	14×2=28	28
4	Long Questions (L)	7×4=28	28
Total		75	75

Note : Third Terminal Examination and Pree SEE Examination questions will be asked according to CDC model question.

First Terminal Examination – 2081
Model Question

Class : 10 F.M.: 50

Subject : Science and Technology Time : 2 Hours

Group A (Multiple Choice Questions) **7×1=7**

Tick the best answer from given alternatives.

1. How many combinations are there in the ASCII code of 8 bits?
   i. 254 ii. 255 iii. 256 iv. 257
2. Pine have cones and naked seed in them. Which sub-division do they belong to?
   i. angiosperm ii. pteridophyta
   ii. gymnosperm iv. bryophyta
3. Which one is the first aid if a person is unconscious due to heart attack?
   i. angiography ii. heart bypass surgery
   ii. ultrasonography iv. CPR
4. Why does a hydrometer float more in seawater than in freshwater?
   i. seawater has a lesser density than freshwater
   ii. freshwater has a lesser mass
   iii. seawater has more density than freshwater
   iv. the hydrometer floats in water
5. Why does diamond spark a lot?
   i. it is transparent ii. its critical angle is 24 degree
   ii. it is hard iv. it is made of glass
6. Which of the following is the most reactive non-metal?
   i. Chlorine ii. Fluorine
   ii. Iodine iv. Bromine
7. Why does phenolphthalein solution turn pink, when ammonia is passed through it?
   i. Ammonia is base, so it can turn phenolphthalein pink.
   ii. Ammonia is acid, so it can turn phenolphthalein pink.
   iii. Ammonia is pink, so it changes everything into pink.
   iv. Ammonia dissolve in water to produce strong acid.

Group B (Very Short Questions) **5×1=5**

Answer the questions in very short.

8. What is the type of variable if its magnitude is constant throughout the experiment?
9. List out any two applications that are used to edit audio.
10. Which type of assisted reproductive technologies can be used for a man who has underdeveloped testicles, weak sperm, sperm count is low or sperms are immotile?
11. A person has a blood pressure of 120/80 mmHg. What does it mean?
12. How much is the specific gravity of gold?

Group C (Short Questions) **9×2=18**

Answer the questions in short.

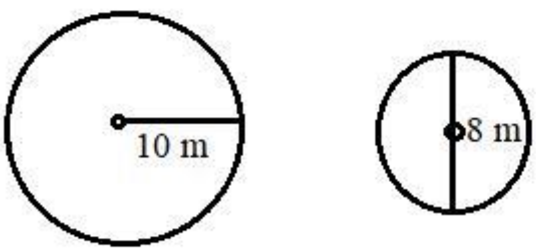
13. Mohan has a cow farm. He increased the food to one group of cow and kept constant for others. The group that ate more food gave more milk than the others. Enlist dependent and independent variables in this experiment along with their reason.

14. Write any two differences between class crustacea and class insecta.
15. Why is spirogyra kept in the division algae? Give any two suitable reasons.
16. Honey bee plays a vital role in pollination, how?
17. The probability of getting hurt is more when jumped from a significant height, why?
18. The mass of the moon is 7.2×10^{22} kg and its radius is 1.7×10^{3} km. Calculate the value of acceleration due to gravity on the surface of the moon.
19. Why does the reactivity of elements increase on moving from top to bottom in group IA of modern periodic table?
20. A reaction between sodium hydroxide and hydrochloric acid is called a neutralization reaction. Show chemical equations along with the reasons.
21. Write any two limitations of the balanced chemical equations.

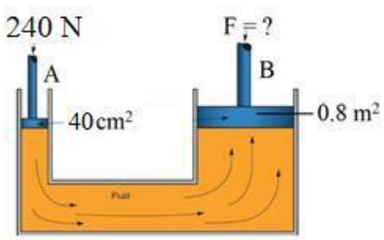
<u>Group D (Long Questions)</u> **5×4=20**

Answer the questions in detail.

22. Based on units, check whether the following equations are correct or not?
    i. $V^2 = \underline{u^2} + 2as$ ii. $S^2 = ut + at^2$
23. Make a table and compare gymnosperm and angiosperm based on their habitat, structure, leaf, flower, fruits and seeds.
24. Mass of large object is 50 kg and that of small object is 20 kg are shown in the diagram. The gravitational force between them is 5×10^{-5} N. If they are separated by 14 m between their surface, what force will be exerted between them?



25. Study the given diagram and answer the following questions.
    i. What is the name of this instrument?
    ii. The cross sectional area of piston A and B is $40cm^2$ and $0.8cm^2$ respectively. Then how much load can be balanced on piston B if 240 N force is applied on Piston A?



26. Study the given tables and answer the following questions.

Li	Be
Na	Mg

i. On what basis, elements are arranged from left to right?
ii. Which one is more active in between Li and Na and why?
iii. Write a formula of a compound made from Mg and Cl.

The End